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A Data Governance Model Based on Multi-Agents System

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***Abstract*—Data assets have become a core driver of societal development; however, data originates from diverse channels and varies significantly in quality, making it unsuitable for direct analysis and decision-making without enhancement through data governance. To address the limitations of traditional data integration techniques in meeting the multi-modal data governance requirements of massive data environments, this paper proposes a data governance model based on multi-agents system. Leveraging the powerful computational and reasoning capabilities of large language Model, this model constructs specialized intelligent agents for tasks including metadata management, data standards management, data integration, data classification and grading, and data quality management, facilitating the evolution of data management from the integration phase towards intelligent governance. The paper further illustrates the model's application within the context of university data governance and concludes with a summary of related work.**

Keywords—data governance, multi-agents, large language model, multi-modal

I. Introduction

With the advancement of digital technologies, many organizations have developed numerous application systems; however, since these systems are often built independently, their internal data dictionaries, structures, and definitions vary, leading to significant challenges in areas such as data quality and data sharing. Centralized data sharing within organizations is currently primarily achieved using Data Middle Platforms, where data governance tasks are typically performed during the data ingestion process [1-2]. However, data synchronization within these platforms predominantly relies on ETL (Extract, Transform, Load) technology, which generally cannot support real-time updates. Furthermore, Data Middle Platforms often fail to adapt promptly to evolving business processes or system upgrades, resulting in disorganized overall data management. Additionally, data collected via these platforms often lacks guaranteed quality due to the absence of effective data quality management mechanisms[3-4].

Data governance has become an important tool for mining the value of data. Data has penetrated into every industry and business functional field today, becoming an important factor of production. Vasanth et al. studied how data governance in education, health care, government and enterprises adopts flexible and extensible frameworks to support data governance, while adopting corresponding strategies to maintain data quality, compliance and interactivity [5-9].

To effectively integrate massive, dynamic and diverse data into valuable information resources, the big data governance model proposed by Wu et al.,which aims to achieve the synergy of human intelligence, artificial intelligence and organizational intelligence with big data supporting data collection, processing, analysis and quality control [10-12].

Alsaad et al. constructed the implementation path of optimizing data governance from the perspective of data life-cycle management, and discussed the path and method of data governance from multiple dimensions such as institutional mechanism, unified data platform, data standards, and team building [13-14].

Most existing data governance methods are based on workflow systems and struggle to address the challenges posed by data complexity in massive data environments, including significant manual effort, data quality issues, and data security concerns[15-17]. The agent-based data governance model proposed herein leverages the cross-domain knowledge and complex task-handling capabilities of large language Model (LLM) and intelligent agents to enhance data governance standards. Aiming to establish a comprehensively integrated, intelligent, agile, and well-governed data enablement ecosystem, this model optimizes governance processes and drives the transformation of data management from unidirectional control towards collaborative governance.

II. METHODOLOGY

A. Data Governance Architecture

The data governance model in this paper aims to solve the core problems in current massive multi-modal data governance from multiple dimensions, including data collection, data quality management, data classification and grading, and data intelligent application. The LLMs is applied to all aspects of data planning, identification, integration and quality improvement, giving full play to the advantages of LLMs in knowledge understanding, semantic reasoning, multi-modal data identification and task scheduling, and realizing the intelligence and automation of data governance. The detailed architecture is shown in “Fig. 1”.

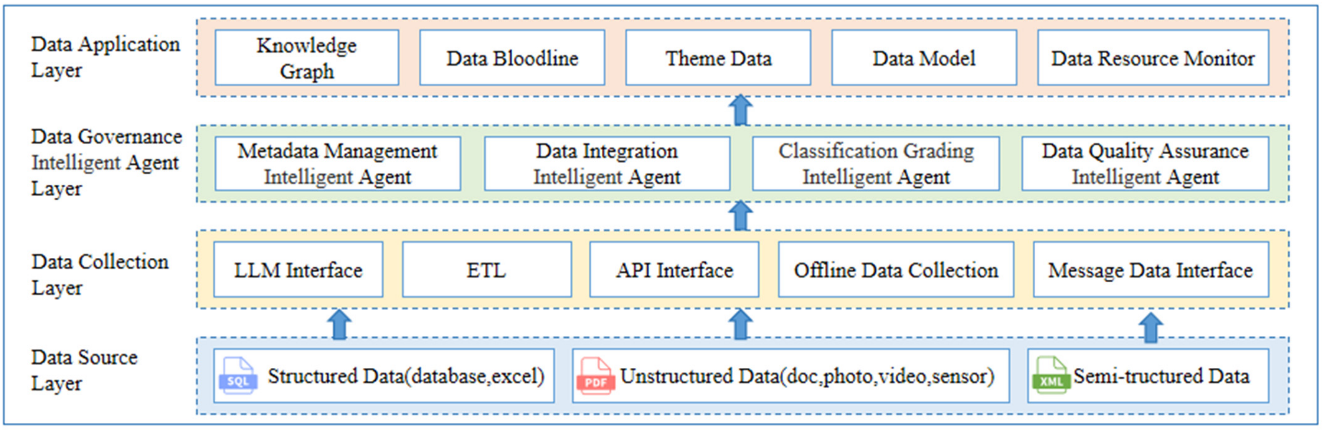

Fig1. Data governance model architecture

The architecture is implemented in a hierarchical design method. Each layer is responsible for the corresponding governance tasks. The data source layer corresponds to a variety of data sources, such as databases, the Internet, multimedia data, and sensor data. The data collection layer provides rich collection tools, such as LLM interface, ETL interface, and API interface message interfaces, to achieve real-time data collection, incremental data collection, and full data collection. The LLMs are responsible for intelligent load balancing decisions for the scheduling services of the collection thread; It supports thread adjustment according to the data volume and scenario requirements. The data governance agent layer is responsible for the automatic data labeling capability, and supports the automatic data cleaning, integration and conversion according to the requirements of relevant rules. This design ensures the flexibility and scalability of the model, and meets the needs of data governance in different scenarios. The data application layer provides data marts and intelligent models, and combines AI algorithm capabilities such as knowledge maps and NLP(Natural Language Processing) to achieve intelligent AI data query[18].

B. Metadata Management

1) Basic Concepts

Metadata is data that describes the attributes of other data, serving to document information about the data itself Metadata encapsulates crucial information such as data source, definition, usage context, and version history. It facilitates the translation between business management terminology and technical terms[19-20]. By utilizing metadata, users can clearly trace data lineage, identify relationships between data assets, and precisely locate data. Analogous to a library’s catalog facilitating book management and retrieval, metadata helps users effectively manage and rapidly locate data.

2) Metadata Identification Agent Model

Metadata includes business metadata, technical metadata and management metadata. Business metadata represents the conditions of data occurrence and use under business rules, technical metadata records the information of data storage and flow, and operational metadata describes the details of operation and access to data. For example,the business metadata of an image has attributes such as name, size, source, and type, technical metadata database, table, and file, and operational metadata attributes such as author, creation, modifier, and authority.

The metadata identification agent utilizes tools provided by LLMs to proactively collect and manage metadata. The metadata identification agent model is shown in “Fig. 2“.

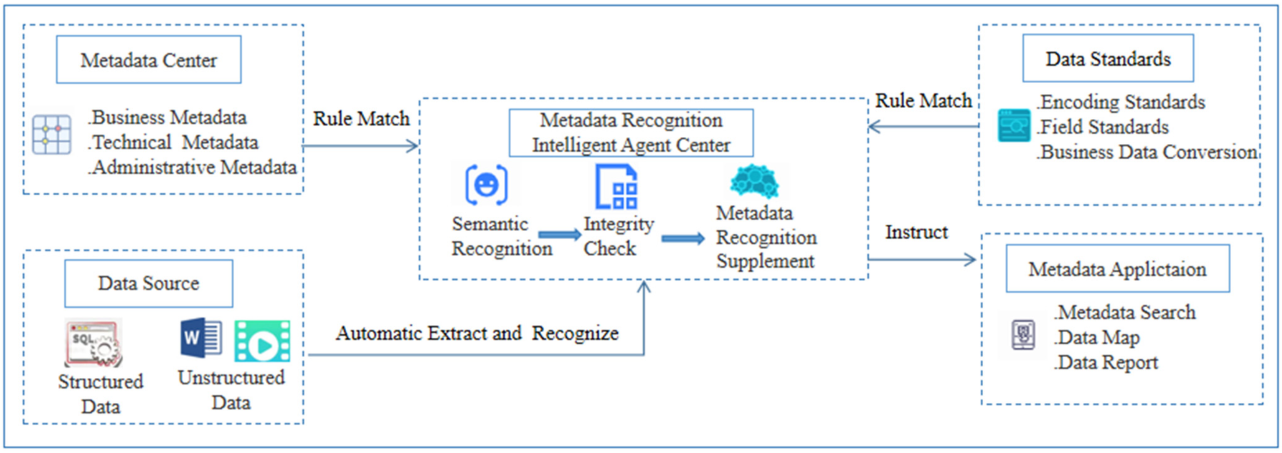

Fig2. Metadata Identification Agent Model

For structured data, the agent's parsing engine automatically identifies database sources, table structures, field comments, and business rules. It extracts information regarding tables, views, stored procedures, functions, packages, triggers, primary/foreign keys, connections, and synonyms to construct technical and management metadata. We can obtain database information through the database Data Definition Language.For instance, it can obtaintable and view information by using the statement "select table_schema, table_name,view_definition from information_schema ".

For unstructured data, the agent employs NLP image recognition, and intelligent summarization techniques to extract and comprehend key semantic information, thereby enriching the metadata. By leveraging contextual analysis, it automatically identifies lineage links to generate a metadata map. Furthermore, the agent can acquire relevant business metadata by analyzing business sample data associated with the data under examination.

C. Data Integration

1) Definition of Data Standards

Data standards are normative constraints that ensure the consistency and accuracy of data usage and exchange[19]. They are fundamental to achieving data trustworthiness, consistency, and correctness. However, standards often differ across disparate sources, leading to issues such as homonyms (same name, different meaning) and synonyms (different names, same meaning). Implementing unified data standards helps standardize data collection, storage, processing, and usage workflows, thereby enhancing overall data quality[21].

By referencing international and domestic standards, data item standards are reviewed and consolidated to establish a standard knowledge base. This knowledge base includes codes for categories such as personnel types, academic majors, organizational units, and business types, typically stored using object-value mapping methods as illustrated in Table 1, the system can reference professional title data conforming to standards knowledge base and standardize "associate Prof.,vice professor, associateprofessor" to "associate professor".

TABLE 1. STANDARD KNOWLEDGE BASE

ID	*object*	*value*
1	professional title	professor
2	professional title	associate professor
3	professional title	lecturer
4	professional title	assistant professor

2) Data Integration Methods

Data integration is the process of consolidating data from multiple sources according to predefined rules, methods, and procedures, enabling centralized storage and management. Data integration methods are typically categorized based on the data source type: integration of structured data and integration of unstructured data.

The integration of structured data typically follows the ETL process, a common methodology in the data integration field. This approach primarily addresses data quality issues such as inconsistencies in code set definitions, data formats, and measurement units. Such issues can typically be resolved through rule-based batch transformation processes, converting the data to comply with established standards. Data quality enhancement is achieved through the application of these ETL transformation rules.

Unstructured data integration involves a multi-step process: ingestion, data cleansing, data transformation and abstraction, data annotation, NLP, and loading. Data cleansing, guided by data standards, involves removing duplicates, imputing missing values, and correcting errors to standardize low-quality data. Data transformation and abstraction convert unstructured data into a structured format while extracting metadata to facilitate analysis and decision-making. Data annotation involves adding labels or tags to data for easier identification and categorization. NLP techniques are applied for semantic analysis and keyword extraction. Finally, loading involves storing the original source data along with the extracted structured information, often in a vector database.

3) Data Integration Agent Model

The Data Integration Agent, particularly in scenarios where data quality issues are identified, leverages the established data standard system to provide automated data cleansing and transformation. The data integration agent model are illustrated in "Fig. 3".

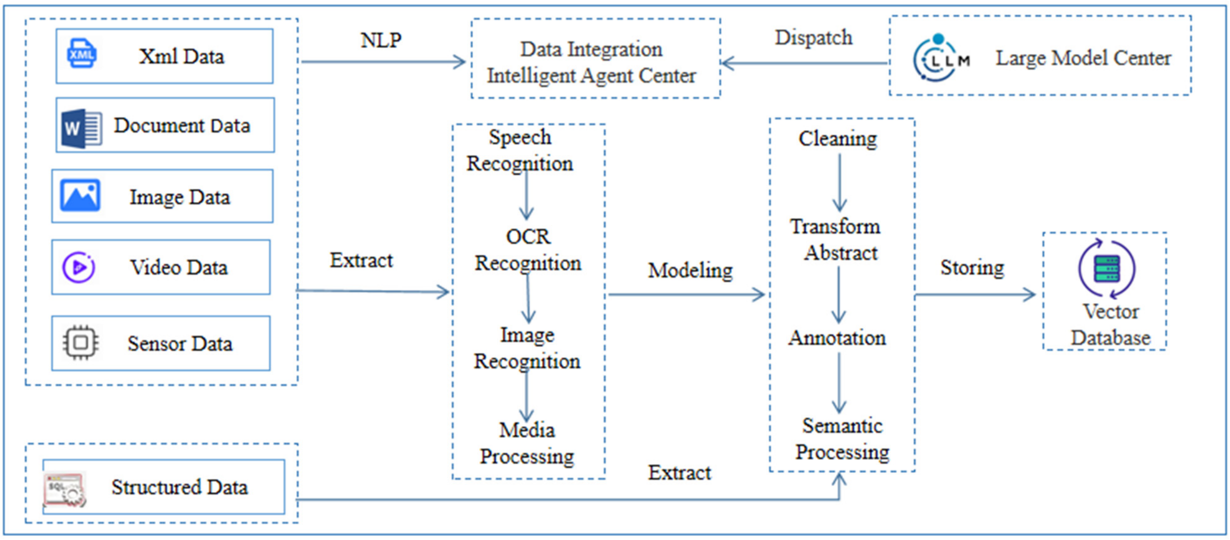

Fig.3. Data Integration Agent Model

During data integration and initialization phases, the agent analyzes data characteristics and interprets data standards to efficiently cleanse and transform data, promoting standardization. Following the cleansing process, it performs quality validation on the standardized results, thereby significantly reducing the need for manual intervention.

Based on the data source type specified for an integration task, the Data Integration Agent employs LLMs powered tools to extract information from unstructured data. It utilizes Optical Character Recognition technology to extract text, mathematical formulas, tables, and charts from images, and employs speech recognition and media processing techniques to extract key information from audio and video files.

The Data Integration Agent can automatically construct integration interfaces tailored to specific data source types. Furthermore, catering to the varying data timeliness requirements of different integration scenarios, it provides capabilities for batch, incremental, and real-time data integration. The agent also manages task scheduling based on defined time windows, supporting serial, parallel, and nested execution patterns. This enables precise control over individual data synchronization tasks, ultimately facilitating an optimized data integration process.

D. Data Classification and Grading

1) Basic Concepts

Data classification involves categorizing data based on its inherent nature and business attributes, establishing corresponding storage structurnd management methodologies; this process should adhere to national and industry classification standards while incorporating the organization's specific business requirements[22-23]. Data grading builds upon classification, assigning levels to data based on its importance and sensitivity, Data is usually divided into public level, common level, sensitive level, important level and secret level according to the impact results of unauthorized operations[24]. The corresponding relationship between data classification and level can be established according to the category of objects. Table 2 shows the security level of personal data.

TABLE 2.THE SECURITY LEVEL OF PERSONAL DATA

level	***title***	***value***
S3	sensitive	Data of personal sensitive information, such as bank card and mobile phone number
S2	common	Data can be mapped to individuals through technical means, such as name and email
S1	public	Personal information can be publicly obtaine, such as title and country

Data classification and grading is typically implemented in the following steps illustrated in "Fig.4":

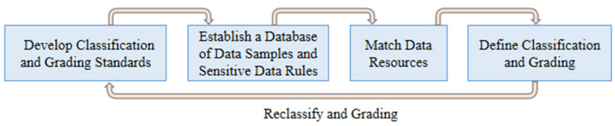

Fig.4.Data Classification and Grading Steps

a) Develop Classification and Grading Standards: Define organizational data classification and grading standards based on applicable national and industry regulations, combined with the organization's specific business characteristics.

b) Create Data Samples and Sensitive Rule Repository: Develop matching rules for sensitive data using techniques like keywords and regular expressions across various data types. Establish data samples and assess the potential impact if data resources (datasets and their items) are tampered with, destroyed, leaked, or illicitly used.

c) Match Data Resources: Identify the datasets and data items within the organization's data resources and apply the rules from the repository to assign appropriate security grades.

d) Review Grading Results: Conduct reviews of the assigned security grades for datasets and data items. If the assigned grade is deemed inappropriate during the review, the data level must be redefined.

e) Data Re-grading: Data re-grading is the process of redefining the security level of data when its content or sensitivity changes over time.

2) Data Classification and Grading Agent Model

The Data Classification and Grading Agent automates the classification and grading process, enhancing both accuracy and efficiency. The operational principle of this agent model is depicted in "Fig. 5".

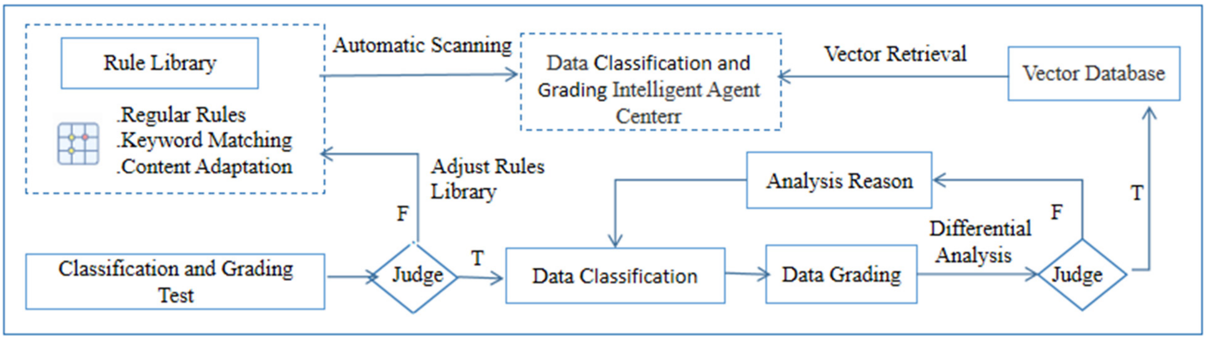

Fig.5 Data Classification and Grading Agent Model

The agent utilizes actual data to automatically test the rule repository. If the results do not align with the classification and grading requirements, the agent adjusts the repository. If the results are satisfactory, the validated rules serve as a template for automated labeling, proceeding first with classification and then grading. Subsequently, the agent analyzes discrepancies in the grading results. If the results meet predefined conditions, the labels are stored in the vector database; otherwise, the data undergoes reclassification.

E. Data Quality Management

1) Definition of Data Quality Management

Data Quality Management (DQM) relies on defining data quality rules and configuring data inspection plans. It utilizes data quality reports to highlight issues and track quality trends over time, establishing an ongoing DQM system aimed at continuous data quality improvement. Data quality can usually be evaluated from the perspectives of data integrity, accuracy, effectiveness, consistency and uniqueness[25-26].

2) Data Quality Management Agent Model

The workflow of data quality management agent model is shown in "Fig. 6",which is realized through the following steps:

a) Formulate data quality inspection rules;

b) Define relevant tables and data items for quality inspection and management;

c) Generate quality template library according to historical quality reports;

d) The LLMs calls the test script to test the quality of the main database, generate a data quality report according to the template, and feed it back to the data authority;

e) The data authority department shall correct the data according to the quality management requirements, and enter the main database after passing the audit.

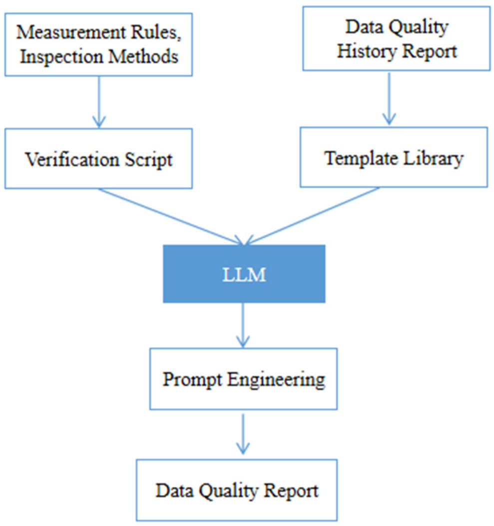

Fig.6 Data Quality Management Agent Model

III. Specific Application Analysis

While most universities have made some progress in data production and integration through the implementation of information systems and data middle platforms, data application and operations often remain in nascent stages. To maximize data value and advance smart campus initiatives, further data governance is crucial. This drives the necessary transition in educational decision-making from experience-based intuition to data-driven insights, and shifts educational services from reactive responses to proactive engagement.

Effective university data governance encompasses the following key aspects:

a) Data Standardization: Surveying and cataloging various data sources across the university, using Ministry of Education industry standards as the primary framework to define data subsets and classifications. This involves establishing internal code standards, coding rules, and naming conventions tailored to the university's specific context. Develop enforcement guidelines for data standards to ensure their consistent application throughout application development and data definition processes, thereby guaranteeing that data output from all systems complies with the defined standards.

b) Metadata Management: Leveraging the LLMs to standardize metadata collection, storage, processing, and usage workflows, thereby enhancing overall data quality.

c) Effective Data Integration: Analyzing, cleansing, transforming, and extracting diverse data types based on varying factors such as application purpose, organizational strategy, and data characteristics to enable cross-system sharing and integration. This includes designing an appropriate vector database architecture to improve data storage, retrieval, and analysis efficiency.

d) Data Classification and Grading Management: Implementing field-level security classification and protection mechanisms based on established data security policies and classification/grading requirements. This involves establishing distinct approval processes based on the security level of the data fields being accessed and supporting the definition and enforcement of data security levels and associated authorizations.

e) Knowledge Graph Construction: Utilizing data maps to provide real-time visibility into departmental data production, sharing, usage (invocation), and alerts. These intuitive visualization tools help managers monitor data dynamics in real time, enabling timely identification and resolution of issues, thereby enhancing the transparency and efficiency of data governance.

f) Data Quality Improvement: When data is exposed via APIs, personalized reports, campus analytics dashboards, or other applications, data consumers can compare the presented information against their own knowledge or context. Upon identifying discrepancies, users can submit feedback through interfaces provided within the front-end applications. This feedback is routed directly to the relevant data source department or provider. A system tracks the issue resolution process, establishing an interactive quality feedback loop between data consumers and providers until the problem is resolved.

IV. CONCLUSION

In the era of big data, data has become a core asset for organizations of all types and a key driver of societal development. Initiatives such as operationalizing data insights and implementing precision marketing fundamentally rely on high-quality data. Data governance systems built leveraging the LLMs and NLP technology focus on the automated and intelligent governance of massive, multi-modal datasets. Through progressively refined data standards and classification/grading rules, these systems enable high-quality automated data classification/grading, data labeling, and intelligent integration, ultimately culminating in a robust, high-quality data resource ecosystem.

Currently, generative AI technologies, exemplified by the LLMs, are rapidly advancing. The practical performance and effectiveness of deployed LLMs heavily depend on the quality and relevance of data within specific industry domains and application scenarios. High-quality data governance significantly enhances the accuracy of model training, thereby providing a solid foundation for subsequent big data analysis and advanced intelligent applications.

REFERENCES

[1] O. Zygmuntowski,"Data Governance in a Trilemma: A Qualitative Analysis of Rights, Values, and Goals in Building Data Commons," Digital Society , pp 1-19, July 2023.

[2] Q. Li, L. Lan, N. Zeng, "A Framework for Big Data Governance to Advance RHINs: A Case Study of China,," IEEE Access ,vol.7 pp 50330 - 50338, April ,2019.

[3] D. Lis, M. Arbter,,Markus Spindler,Boris Otto,"An Investigation of Antecedents for Data Governance Adoption in the Rail Industry—Findings From a Case Study at Thales," IEEE Transactions on Engineering Management Volume: 70, pp.2528 - 2545 Issue: 7, July 2023.

[4] F. D. A. Vilela,R. R. Ciferri,"A novel solution to perform real-time ETL process based on non-intrusive and reactive concepts,"2021 International Conference on Computational Science and Computational Intelligence (CSCI), pp.556-561,December 2021.

[5] Vasanth,Saravanan, Keerthana,M. Karthick,"Data Governance in the Big Data Era: A Scalable Framework for Effective Management and Regulatory Compliance,"2024 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES),December2024.

[6] S. O. R, P. W. Handayani and A. N. Hidayanto, "Health Data Governance Issues in Healthcare Facilities: Perspective of Hospital Management," 2022 10th International Conference on Information and Communication Technology (ICoICT), pp. 1-5,2022.

[7] K. Bugbee et al., "Enabling Dynamic Data Governance in Science: Design, Implementation, and Future Directions of the Modern Data Governance Framework," IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium, pp. 3720-3723, 2024

[8] R. Budiarto, I. K. S. Buana, H. Kabetta, H. Setiawan, Girinoto and N. Qomariasih, "Enhancing Data Governance and Personal Data Protection: A Strategic Review for Cases in Indonesia," 2024 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS), pp. 625-630,2024.

[9] H. Y. Kim and J. -S. Cho, "Data Governance Framework for Big Data Implementation with a Case of Korea," 2017 IEEE International Congress on Big Data (BigData Congress), pp. 384-391,2017.

[10] H. Jiang and W. Yan, "The Application of Data Governance in Universities," 2023 6th International Conference on Artificial Intelligence and Big Data (ICAIBD), pp. 54-59, 2023.

[11] X. Wu,B. Dong,X. Du,W. Yang,"Data Governance Technology," Journal of Software,Vol.30(9),pp.2830−2856,2019.

[12] V. S, G. Saravanan, K. S and M. K. A, "Data Governance in the Big Data Era: A Scalable Framework for Effective Management and Regulatory Compliance," 2024 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES), pp. 1-5,2024.

[13] A. Alsaad, "Governmental Data Governance Frameworks: A Systematic Literature Review," 2023 International Conference on Computing, Electronics & Communications Engineering (iCCECE), pp. 150-156, 2023.

[14] F. R. Hendrawan, T. F. Kusumasari and R. Fauzi, "Analysis of Design Implementation Guidelines for Data Governance Management Based on

DAMA-DMBOKv2," 2022 Seventh International Conference on Informatics and Computing (ICIC), pp. 1-6), 2022.

[15] B. Guan, J. Cao, X. Wang, Z. Wang, M. Sui and Z. Wang, "Integrated Method of Deep Learning and Large Language Model in Speech Recognition," 2024 IEEE 7th International Conference on Electronic Information and Communication Technology (ICEICT), pp. 487-490,2024.

[16] I. Trummer, "Large Language Models: Principles and Practice," 2024 IEEE 40th International Conference on Data Engineering (ICDE), pp. 5354-5357, 2024.

[17] E. Verhelst and T. Belpaeme, "Large Language Models Cover for Speech Recognition Mistakes: Evaluating Conversational AI for Second Language Learners," 2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI), pp. 1705-1709, 2025.

[18] I. Song and K. Lee, "Optimizing Communication and Performance in Federated Learning for Large Language Models," 2025 International Conference on Artificial Intelligence in Information and Communication (ICAIIC), Fukuoka, pp. 0964-0967, 2025.

[19] O. Azeroual, A. Nikiforova and K. Sha, "Overlooked Aspects of Data Governance: Workflow Framework For Enterprise Data Deduplication," 2023 International Conference on Intelligent Computing, Communication, Networking and Services (ICCNS), pp. 65-73, 2023.

[20] J. Xie and J. Chen, "Text-to-image Retrieval Based on Zero-shot Transfer Learning with CLIP Model and Vector Database," 2024 11th International Conference on Behavioural and Social Computing (BESC), pp. 1-7, 2024.

[21] Aneja, A., Dhull, A., Singh, A., Singh, “Large Language Model on Multi-Modal Data. In: Singh, A., Singh, K.K. (eds) multi-modal Generative AI," Springer, pp.63-77,February 2025.

[22] Naguib, H.M. Kassem, H.M. & Naem, “The impact of IT governance and data governance on financial and non-financial performance," Future Business work, Vol.10 ,pp.3-22, January 2024.

[23] Tamizshelvan, C.Vijayalakshmi,“Cloud data access governance and data security using distributed infrastructure with hybrid machine learning architectures," Wireless Netw,Vol. 30, pp.2099–2114 ,January 2024.

[24] de Pedraza, P., Vollbracht, “General theory of data, artificial intelligence and governance," Humanit Soc Sci ,Vo 10, pp.1-16, September 2023.

[25] K. Chen, Y. Yang, B. Chen, J. A. Hernández López, G. Mussbacher and D. Varró, "Automated Domain Modeling with Large Language Models: A Comparative Study," 2023 ACM/IEEE 26th International Conference on Model Driven Engineering Languages and Systems (MODELS), pp. 162-172, 2023.

[26] M. Ji, X. Gu, Q. Guo and X. Ding, “Research on Government Data Governance in the Era of Large Language Model," 2024 IEEE 9th International Conference on Data Science in Cyberspace (DSC), pp. 668-671, 2024.